

Recent Advances in Rotor Design of Vertical Axis Wind Turbines

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ABSTRACT

The following work represents the most recent advances in design and testing of vertical axis wind turbines (VAWT) rotors. VAWTs have received much attention as of late due to proposed advantages in small scale and off grid wind power generation. Thus, many recent works have surfaced involving analysis, design and optimization of VAWT rotors in order to more efficiently convert wind energy to electricity or other readily usable means. This paper is a collection of most of the recent literature works involving VAWT rotor design and testing, the majority of which published after 2005. We discuss research in the designing of various lift based rotors as well as some drag based rotors, hybrids, and various others. The recent work in this area suggests VAWT capacity could dramatically increase in the near future, and play a vital role in obtaining cleaner, more sustainable energy when global energy demand is increasing at an unprecedented rate.

HIGHLIGHTS

- A review of various works involving rotor design and testing of both lift and drag Vertical Axis Wind Turbines (VAWTs) is presented;
- Benefits of vertical axis wind turbines in small scale and off grid wind power generation is summarized;
- Much of the recent work, published after 2005, has been directed towards analyzing, designing, and optimizing rotor shapes.

The body of this recent work suggests that research on VAWT rotor design continues to flourish and could make VAWTs a viable competitor to more traditional Horizontal Axis Wind Turbines (HAWTs) worldwide.

Keywords: wind, energy, vertical axis wind turbine, VAWT, Darrieus, Savonius

1. INTRODUCTION

Wind turbines have been used by humans for centuries, perhaps even for millennia. One of the earliest historical mentions of windmills, used for the purpose of pumping water, dates as far back as 400 BC [1]. However, direct evidence of the use of windmills or wind turbines for energy conversion was not reported until the 13th century AD when the first drawing of a Persian windmill for grinding grain was recovered [1]. Over the next several centuries, windmills of the horizontal axis windmill design gained popularity for water pumping and

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grinding grain, most notably in northern Europe. Although wind turbines have been used to generate electricity for some time, it wasn't until the mid 1970s that production of reliable, sizable wind turbines could be produced for the masses [2].

Since then, wind turbines gradually increased in size until today, with larger designs topping 6MW capacity with rotor diameters of over 125 m [3]. These larger turbines typically rotate around a horizontal axis and have been in development for some time, with design practices well known to manufacturers. What is generally not well-known is that wind turbines have two main types, referring to their axis of rotation: the Horizontal Axis Wind Turbine (HAWT) and the Vertical Axis Wind Turbine (VAWT). The HAWT rotates along an axis horizontal to the ground, while the VAWT rotates around an axis vertical to the ground, Fig. 1. By far, HAWTs have received the most attention in industry and practice, and the science of designing such rotors is now well-established. In the early stages of wind turbine design, both HAWT and VAWT types received similar levels of interest, but structural and bearing loading issues in VAWT design pushed most early researchers to vacate this field. As a result, comparatively little has been invested in the vertical axis turbine until recently, when applications in off grid electricity generation and small scale power supply renewed interest in this technology.

The reason for the rise in popularity of the VAWT is that generally they emit lower acoustic emissions and have better responses to varying, unsteady and turbulent wind flows [4]. This makes them more suitable for urban areas, as well as lower power and off-grid production close to power demand. Problems still persist in dealing with oscillating loading of the support and ensuring self-starting, especially in larger turbines, where more research attention is warranted. One of the most attractive features of the VAWT is the placement of the generator/gearbox: in HAWTs it is housed in the nacelle, high up in the air, making maintenance difficult and expensive, whereas in the VAWT it can be located conveniently on the ground. Another advantage of VAWTs is their omni-directional nature, as they accept wind from all directions equally. This feature, coupled with a self-starting nature (in some designs), give the VAWT a feasibility advantage over HAWTs in small urban settings.

For these and other reasons, the recent literature is populated with many recent studies which investigate the most efficient and compact VAWT designs for urban electricity production, among other uses. This paper is a summary of the more recent works related to VAWT rotor design, and is limited to scholarly publications since 2005, with a few exceptions.

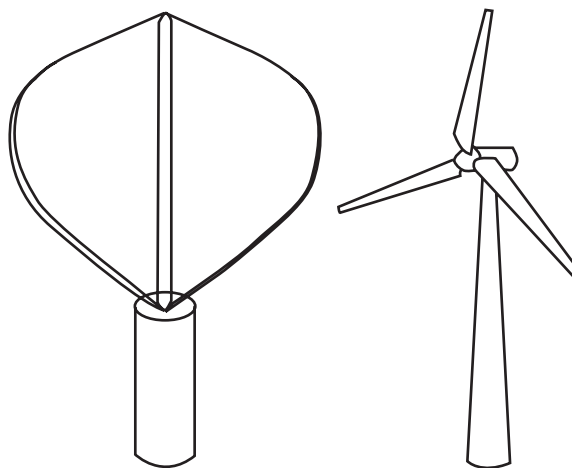


Figure 1: Vertical (left) and horizontal (right) axis wind turbines.

Notable VAWT manufacturers are not mentioned unless as part of a scholarly article, and only the most notable canonical patents are acknowledged.

The most recent review article on VAWTs by Bhutta et al [5] provides a thorough recap of the general VAWT types and configurations, as well as a good perspective on the difference between HAWT and VAWT designs. Eriksson et al [6] assess the various design concepts related to wind power. A more succinct historical perspective of Wind Energy Conversion (WEC) is presented by Sorensen [1].

Although this review discusses only the most recent advances in design and testing of VAWT rotors, the reader will notice a considerable amount of works involving Computational Fluid Dynamics (CFD) approaches, which provide a much cheaper alternative to scale modeling and testing. Until most recently, the most widely used models to investigate wind turbine performance have been based on blade element theory and/or potential flow modeling, and included stream tube, vortex and cascaded models [53]. Blade element theory considers turbine blades as a series of two dimensional cross sections, and assumes they act independently of each other [48]. Until recently, this type of analysis is still used in HAWT simulation; however problems arise when considering VAWTs, when upstream effects must be taken into account. For example, in Darrieus type rotors, downstream blades must pass through the wake of upstream blades, rendering the aforementioned blade element theory impractical.

Due to recent increases in computing power, more complex models have been developed which include sophisticated turbulence modeling, which can provide better insight into VAWT rotor behavior. More importantly, these turbulence models have shown to have a great impact on the accuracy of torque prediction [67]. A more comprehensive and recent review of wind turbine aerodynamics, including CFD and modeling used to estimate rotor and support loading is offered by Hansen and Madsen [7]. For information regarding rotor control for electricity generating purposes, other recent reviews (e.g. [8]) are available.

The remainder of this review is divided into four sections. The first two are based on research concerning each type of VAWT, namely drag and then lift-based designs. The final two sections deal with hybrid designs and recent experimental, offshore, or other works furthering the science of VAWT design.

In order to more succinctly compare recent designs and studies, we must first introduce the following non-dimensional entities widely used in wind turbine design: the *power coefficient*, which is synonymous with the efficiency, is defined as the ratio of the power obtained to the power available in the wind; the *torque coefficient* is a non-dimensional representation of the rotor torque, which is proportional to the power produced; the *solidity* is a measure of the area occupied by the VAWT, and is defined as the blade face area divided by the rotor swept area; finally, the *tip-speed ratio* is defined as the ratio of the blade tip speed to the free-stream wind velocity.

2. LIFT-BASED VAWT DESIGNS

Lift based VAWTs are designed such that resultant forces perpendicular to the local airfoil wind velocity induce rotor torque. That is, the lift force is primarily used to elicit motion of the turbine rotor. The majority of early windmills were in fact lift-based, and it wasn't until Darrieus' patent in 1931 [9] that lift based wind turbines were considered viable for electricity production. This patent covered almost all conceivable arrangements using vertical airfoils on a rotating shaft and thus gave rise to H-type, Gyromill and Egg-Beater design types, Fig. 2; today almost all lift based rotors are based on this original patent. There are also many other rotor shapes possible, including "V", " Δ ", and "diamond" shapes [49], however in practice those shapes shown in Fig. 2 are by far the most common.

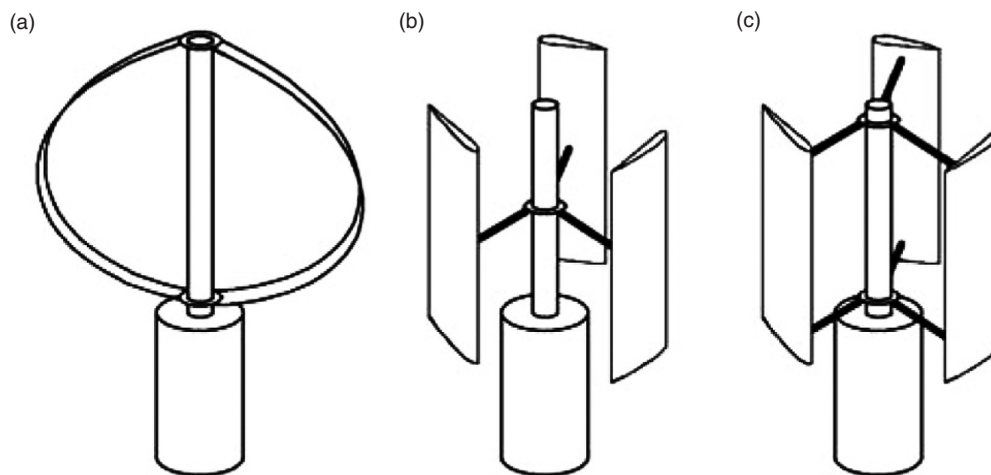


Figure 2: (a) Egg-Beater, (b) H-type and (c) Gyromill VAWT rotors.

Whereas many VAWTs use fixed geometry rotors, some interesting endeavors are dedicated to the active control of pitch angles of the airfoils themselves. This subject is fairly well researched in itself [10] and provides a good starting point. In terms of rotor design, many recent studies have been conducted involving individual blade pitching techniques. Of particular interest to VAWT designers are passive control mechanisms, which pitch blades according to the materials in the turbine itself, and differ from active control strategies in that no actuators or sensors are needed. The result is greatly reduced costs associated with actuators or sensors otherwise required for active control mechanisms. As an example, Liu et al [64] experimentally test a maglev (magnetic levitation) rotor outdoors, intended to increase performance by including blade pitching as well as greatly reducing rotor bearing friction. The blade pitching mechanism is passive in nature, and operates via a hinge constrained on each side. The maglev apparatus was reported to enjoy lower startup wind speeds, higher power coefficient, lower noise emissions and longer lifetime than alternative VAWT designs. The design type is already in use in Shandong University in China [65] where a test apparatus powers a streetlight. As an example of active pitch control, a recent study proposed by Hwang et al. [58] analyzes the improvements in efficiency possible through a four-bladed VAWT rotor employing a control strategy utilizing a small micro-controller and several sensors and motors. Their method generates around 30% more power than with a fixed blade approach, giving more plausibility to VAWT blade pitching, both active and passive.

Though blade pitching remains an important aspect of VAWT design, one of the more pertinent design challenges remains in addressing, improving, or enhancing the self-starting of rotors. Self-starting is not guaranteed by any VAWT rotor in all conditions, and depends heavily on many factors including solidity, rotor weight and wind conditions. CFD techniques are used by many to investigate this phenomenon, including by Hill et al [13] and Dominy et al [14] to analyze the parameters affecting the self-starting characteristics of an H-type rotor. Using computational techniques and a NACA 0012 airfoil, it is concluded in [14] that in some steady wind conditions, a three-bladed, lightly loaded H-type Darrieus VAWT always has the ability to self start, whereas a two bladed VAWT may only self-start under favorable conditions. Although these results seem promising, a caveat is added by Wood et al [15], who investigate how well CFD can predict, if at all, self-starting. Using CFD techniques and relating to experimental data, their findings indicate that while computational means provide reasonable insight into the steady operation of VAWTs, they often are of poor reliability during

simulation of the startup period, and do not accurately represent phenomena seen in experiments, such as the initial plateau of angular velocity, before the maximum constant final rotor speed is reached [15].

Attempts have also been made recently to enhance startup using passive blade mechanisms. A novel design introduced in Qu et al [16] envisions a Darrieus straight bladed VAWT with the added feature that the blades can open and close along its length via a hinge. During startup, the blades open in the required position along the rotation, and induce drag which causes startup. The authors conclude that the resulting self-adapting, self-starting turbine has greater starting performance than a similarly designed rigid bladed Darrieus rotor.

By a wide margin, most VAWT research works are concerned with rotors employing airfoils with symmetric cross-sections. However, Beri and Yao [17] perform one of the few recent studies in the literature investigating a Darrieus rotor with asymmetric blades. The effects of using an asymmetric profile on self-starting are investigated using a two dimensional CFD analysis on a three-bladed system utilizing NACA2415 profiles. This analysis concluded that while a cambered airfoil can have greater potential to self-start, there are losses encountered resulting in a lower performance at operational speeds associated with higher tip-speed ratios. The same authors perform another similar work in [66] where a modified NACA0018 airfoil with a hinged tail is shown to have better self-starting performance than the NACA 2415.

An experimental study involving asymmetric airfoils was conducted by Takao et al. [54], where the effect of guide vanes on a Darrieus straight bladed VAWT is analyzed. Building upon a previous study by Takao et al. [55] and Kuma et al. [56], where directed guide vanes were shown to be advantageous in straight bladed Darrieus VAWTs, the study in [54] uses a rotor with NACA4518 airfoils and experimentally investigates the effect of various arc plates configurations used as guide vanes. Experiments were conducted with vane configurations using 1, 2, 3, and 4 arcs, and an optimum spacing of vanes is determined. Interestingly, it was also found that the power and torque coefficients were independent of the number of guide vanes used.

Since VAWTs are well suited to low power, off-grid production, one of the more popular applications of lift-based VAWT designs is for use in urban areas, more specifically on the rooftops of buildings. Rooftop mounting can be advantageous in certain conditions, and Balduzzi et al [4] investigate numerically the flow dynamics of various building shapes in proposed urban locations. Using the flow fields obtained, a second CFD analysis is performed, simulating an H-Darrieus rotor subjected to a skewed velocity profile due to the induced boundary layer from nearby buildings. The results indicate that whenever a VAWT is installed atop a building notably higher than surrounding constructions, significant increases (up to 70%) in capacity factor can be observed. Conversely, if nearby construction objects are higher than the building, capacity is lessened. As a result, an increasing number of manufacturers are now designing VAWTs intended for rooftop installation, including for example the Turby turbine [18], a helical shaped Darrieus VAWT.

Although CFD analyses are still quite popular in VAWT analyses, nonetheless there have been notable experimental studies as well, which can be combined with CFD analyses to increase the impact of research. In one such combined study, McLaren et al [19] investigate a high solidity VAWT alongside an equivalent experimental apparatus. The H-type Darrieus rotor is physically built and simulated numerically, using Reynolds-Averaged Navier Stokes (RANS) turbulence modeling. The results indicate that a higher-solidity rotor results in more significant wake effects from upstream blades than would otherwise be realized with a low

solidity turbine. McLaren et al [19] also investigate the vibrational behavior of the same experimental apparatus in [20], and determine that the primary vibration response was due to the rotation of the turbine itself - in other words, generally coinciding with the rotor rotational frequency. One interesting finding was that there can be significant resonance behavior in the struts that support the straight bladed airfoils, and these vibrations change dramatically when flow conditions change.

Most experimental and numerical works in recent times are concerned with traditional blade designs such as those shown in Fig. 2. However, some novel applications of lift-based VAWTs have been developed using a multi-tier (staging) system. For example, a two-stage Darrieus rotor was developed by Gorelov and Krivospitsky, [21]. Two configurations were designed, one with six blades - three top, three bottom - and one with four blades - two top, and two bottom. The staggering of airfoils in subsequent armatures is intended to aid in the self-starting of the rotor. Experimental tests were performed in 1 kW and 3 kW setups, with maximum efficiencies reaching 36% in the 1 kW apparatus and 40% in the 3 kW design.

Rotor designs of VAWTs are not limited to strictly vertical airfoil orientation; indeed many manufacturers design rotors with canted, curved, and helical blades, as well as the traditional straight-bladed design, Fig. 3. Helically twisted turbine blades are sometimes attributed to Gorelov [50] who patented his design for use in flowing water. For example, [22] analyzes H-Darrieus turbines using straight blades, canted blades, curved blades, and canted blades with fences. Airfoil fencing has been used in aircraft wing design for some time, and serves to reduce airfoil stall on wings, since flow and thus separation effects will travel transversely down the wing of a swept-wing aircraft. The experiments in this case [22] indicate that rotors with canted blades experience less local flow reversal compared to the straight bladed design, and the fencing of blades increased power while reducing the tip-speed ratio where peak power occurred.

One of the major attractions of VAWTs is their omni-directionality. In other words, they do not need to be oriented in any particular manner with respect to the wind, and operate equally well regardless of wind direction. For this reason, they are well suited to unsteady wind conditions where similarly sized HAWTs may not achieve good performance. In light of this, recent studies by Scheurich and Brown [23], and Scheurich et al [24], analyze the aerodynamics of various rotor configurations in unsteady wind conditions using a CFD approach employing a vorticity transport model. Three rotor shapes are tested: straight, curved, and helically twisted (for example, conceptually similar to that in Fig. 3) to assess rotor performance in both unsteady and steady wind conditions. In constant rotational speed, the helically twisted blades are shown to be somewhat more efficient in unsteady wind

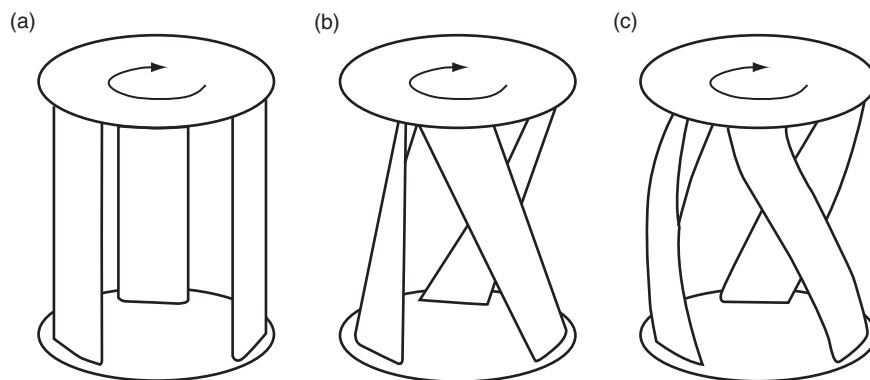


Figure 3: (a) Helical, (b) straight-bladed and (c) canted Darrieus gyromill type VAWT rotors.

conditions. For steady wind conditions, the straight and curved bladed rotors experienced considerable variations in torque when compared to the helical blades, indicating that the helical blade configuration could possibly be better suited to avoid oscillatory aerodynamic loading. Another related work concerned with variable wind velocity in a Darrieus type rotor is performed by Wang et al [25]. Using RANS modeling to simulate airfoil and wake dispersion, it was determined that when increasing velocity while holding rotor speed constant, the total torque coefficient tended to plateau.

3. DRAG-BASED VAWT DESIGNS

Drag-based turbines are conceptually the simplest VAWT design type. In this type, the resultant airfoil force coincident with local wind velocity is used to drive the rotor. Most are modeled after the original Savonius turbine, developed in 1922 [6], and consist of semi-circular scoops in two or three blade configurations, Fig. 4. A drag-type device, it extracts energy from the wind by virtue of the difference in drag experienced by each blade. Because of the shape or curvature of each blade, one side of the rotor experiences more drag when moving away (retreating) from the wind, and less drag when moving towards the wind (advancing), enabling the rotor to spin. Increasing wind velocity and/or aspect ratio in general improves the power coefficient of Savonius turbines [51], in some cases to a maximum of 31% [47]. However, since Savonius rotors cover the equivalent of the swept area in blade construction, their weight to power ratio is quite high, making them inefficient for all but small power generation. For example, Wilson and Lissaman [47] state that a Savonius rotor requires 30 times more surface to achieve the same power as a conventional HAWT wind turbine rotor.

In general, lift-based VAWT rotors are more suitable for higher rotational speeds and will produce more power when compared to drag-based VAWT designs. Since most drag based rotors are slower rotating and may not be as capable of high power production, they are typically unsuitable for large or mid-scale power conversion. They can however still serve many purposes such as lower power production, off-grid electricity production or other uses. For example, a drag-based VAWT, coupled to a stirring vessel is investigated by Ghatage and Joshi [26]. Since electromechanical systems can be expensive and inefficient, this particular design is developed not for generating electricity but for stirring, i.e. for mixing in the process

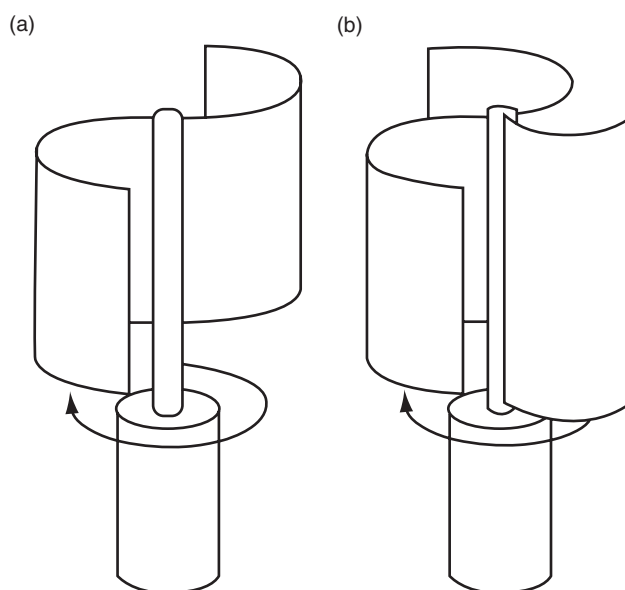


Figure 4: Savonius drag-type rotors, with (a) two blade and (b) three blade configurations.

industry. Through design and testing of a twisted drag-based rotor, the twisted two-bladed rotor was shown to have higher efficiency than the three-bladed design for this stirring application. In closer examination of the two-bladed rotor of [26], an optimum twist angle for the drag-based rotor used for stirring was found to be 30° .

In another example of the off-grid generating capability of drag-based VAWTS, a novel Savonius-type wind turbine was developed recently [27] which uses a series of stacked rotors in order to power an off-grid cellular communications tower. The apparatus involves a stacked turbine installed along the length of the tower in order to provide power for the tower itself. Improvements to traditional bucket-style Savonius blades are made, including venting apertures in the blades themselves (see Fig. 5) to reduce negative drag, as well as the use of caps on top and bottom of the blades. Significant improvements in performance are found using these changes and the design is found to be viable for its intended purposes. In another study assessing caps or plates fitted on the rotor, [28] finds that a spiral Savonius rotor with a middle plate installed halfway up the rotor results in higher and more steady torque during operation.

The optimum configuration of Savonius rotors in one, two, and three bladed systems were investigated by [30] experimentally, using a wind tunnel. In addition, the number of stages is also studied, testing configurations with one, two, and three stacked rotors, all with semi-circular blades. The conclusion through testing indicates that regardless of the number of stages, the optimal number of blades is two. The optimal number of stages was interestingly also found to be two. Both semi-circular and twisted bladed geometries were also tested in [30]; findings indicated that the twisted blade geometry generally performed better than the more traditional semi-circular geometry. Vents are also investigated (Fig. 5) to reduce negative blade drag, and corroborating the findings of Abraham et al [27], it is found that these vents do in fact increase performance in the three-bladed rotor.

Drag-based and Savonius rotors are not limited to the more popular and simpler semi-circular blade design. Indeed, a multitude of designs have risen with creative ways to increase drag on the retreating side while reducing drag on the advancing side of the rotor. For example, McTavish et al [29] envision a novel rotor with many highly asymmetric vertically stacked drag-based rotors. This unique rotor was tested using CFD techniques, and it was found that the static torque generated in this setting is similar to that of existing Savonius

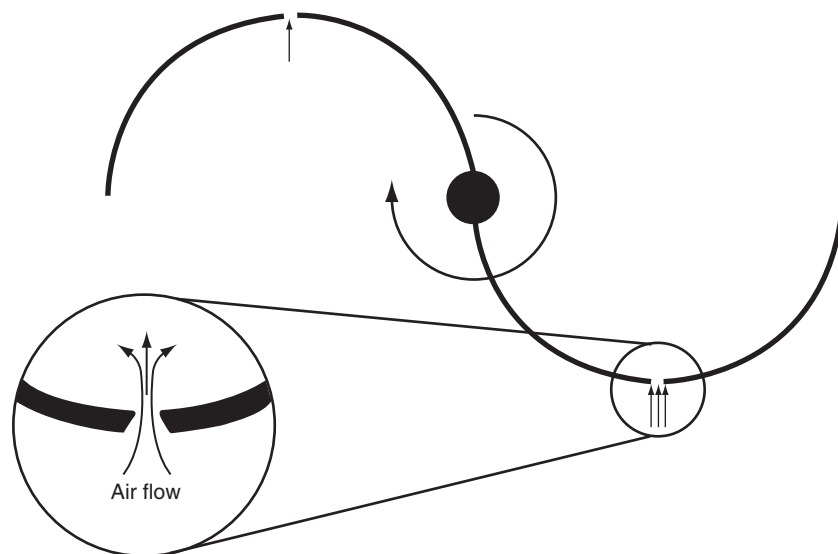


Figure 5: Two bladed Savonius rotor showing directional vents.

rotor. However, the dynamic torque decays more rapidly as rotor speed increases due to the asymmetric design of the turbine. Another example of a unique drag-based rotor is the Wollongong type turbine, investigated by Wang et al [11], and previously analyzed by Cooper and Kennedy [12]. This drag-based rotor is actually shaped much like the Darrieus gyromill turbine, Fig. 2c. However, instead of lift providing the rotor torque, the airfoils are rotated through a bevel gear pair on the top of each blade, ensuring high drag during blade retreat and low drag when blade velocity is opposite the wind. This mechanism uses only the resultant forces on each airfoil to pitch other airfoils, and thus a costly control system is avoided in lieu of a simple pair of gears. The aforementioned studies have shown it to be effective mainly in low tip-speed ratios. Research continues involving this turbine to determine the optimal gear ratio so as to have almost parallel blade with respect to the oncoming wind during blade retreat, and perpendicular to the oncoming wind during blade advancing [11].

Another improvement on the Savonius rotor is investigated by Altan and Atilgan [31], where the use of curtain vanes are implemented in order to direct flow into the retreating side of the rotor. These curtains also serve to reduce the amount of wind striking the negative drag (or convex) portion of the turbine blade. The authors show that the use of curtains can increase performance, with longer curtains attaining better performance than shorter ones. Similar to the concept of curtains, [32] involves an improvement of the Sistan type turbine, one of the oldest wind turbines known. The Sistan turbine operates much in the same way as the Savonius turbine, but with straight blades and with a wall shielding the advancing side of the rotor, reducing negative drag. Through utilizing a wing-wall (Fig. 6) and installing the rotor between discs or caps, Muller et al [32] discuss the use of this type of turbine for use on buildings, and claim high efficiencies can be achieved using these means.

Recently, a different type of drag-based rotor concept, called a Zephyr turbine, has been tested both experimentally and numerically [33, 34] and [35]. The Zephyr turbine is similar to the Savonius turbine in the sense that the rotor involves curved blades; however also present in the Zephyr system is an outer stator which directs flow into the rotor (Fig. 7). This VAWT is analyzed by Pope et al [34] to determine the effects of different stator vanes on the turbine performance, including vanes with and without tabs. Using CFD techniques and RANS

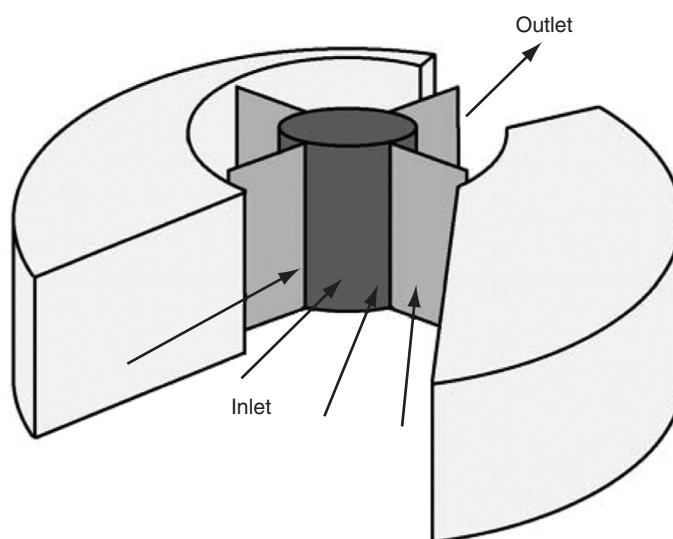


Figure 6: Wing-wall turbine for use on buildings, modified from [32].

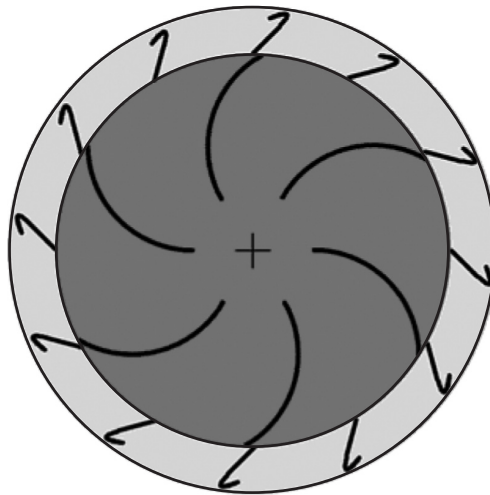


Figure 7: Zephyr turbine, with 12 stator and 6 rotor vanes, similar to that in [33].

turbulence modeling, as well as experimental analysis, an optimal tip-speed ratio is determined for this configuration of turbine. Similarly, Pope et al use CFD techniques to develop a new predictive model involving turbine geometry and dimensional analysis, which may prove useful in other drag-based VAWT designs, [35]. In another case Pope et al also investigate the energy and exergy efficiencies of four turbines; two HAWT types and two VAWT types, one a Savonius and one of the Zephyr types [33]. Through both energy and exergy analysis, it was concluded that second law (exergy) analysis provides a more useful tool to compare the performance of these wind turbines. Furthermore, that the exergy analysis should prove more useful in comparing wind turbines of all types.

4. HYBRID DESIGNS

Hybrid type VAWTs combine two or more different rotors in the same turbine (Fig. 8). Many attempts have been made to combine the lift-based, high power Darrieus type rotors with the high torque drag-based Savonius types. For example, Gavalda et al [36] investigate both a Savonius and a Darrieus type turbine experimentally, as well as postulate the combining of the two into a Darrieus-Savonius turbine where the central part, near the rotational axis, is occupied by the Savonius type turbine. This arrangement can enjoy higher starting torque than that experienced by only the Darrieus rotor. The outstretched armatures contain straight bladed airfoils, and this combination may serve to attain a higher power coefficient than in either system alone in varying wind conditions.

More recently, Kou et al [37] combine a two-tiered Savonius rotor with a three-bladed Darrieus gyromill rotor. Through use of the Savonius rotor, the combined VAWT is shown through experimental means to enjoy a lower self-starting wind speed and better conversion efficiency than without the addition of the Savonius rotor. Alam and Iqbal [38] conduct another study of a Savonius-Darrieus combined rotor. Here, a four-bladed Darrieus rotor is placed atop a Savonius two-bladed rotor. Again, the self-starting nature of the Savonius rotor works well with the higher efficiency (at higher wind velocities) Darrieus rotor. Although this apparatus is tested in flowing water for hydroelectric purposes, it is indicated that the findings also apply for turbines of this type in WEC applications.

The egg-beater type Darrieus rotor has also been combined with Savonius rotors, as in [39] by Gupta et al. Here, the three bladed Darrieus rotor is combined with a three-bucket

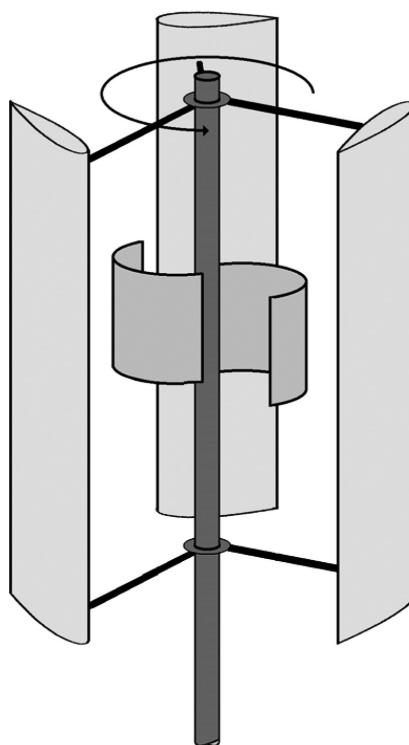


Figure 8: A typical hybrid VAWT rotor.

Savonius rotor. The drag-based rotor is modified to have varying overlapping in blade geometry; that is, the blades are staggered so that they do not meet at the axis of rotation, rather a prescribed distance away from it (for example, Fig. 4 shows blades with no overlap). The conclusion in this work is that performance begins to decrease when rotor overlap is sufficiently increased, and in this hybrid design, maximum performance occurred with no overlap. This work, as in most other reported sources in this document and elsewhere, indicate that hybrid VAWTs are better suited than Savonius only rotors in terms of both power coefficient and energy conversion potential.

5. RECENT EXPERIMENTAL TESTING AND OTHER VAWT RELATED WORKS

The science of wind turbine design has historically progressed through the use of experimental design. It is only over the past 20 years that computational techniques have advanced to the point of viable importance, due to the decrease in computing costs and through the development of CFD analysis. Though CFD techniques are continually improving, there are still significant improvements that must be made in order to accurately represent the complex time and space dependent flow fields associated with VAWT simulation. For this reason, experimental design of VAWTs are still viable, and in fact, many of the aforementioned works involve experimental testing of rotors to determine the effects of rotor geometry on torque and power coefficient. For this reason, Deng et al [40] develop a simple straight-flow wind tunnel test apparatus in order to investigate VAWT performance. The results should prove useful for those wishing to build a wind tunnel apparatus for the purposing of testing a VAWT rotor. Wind tunnel blockage effects can also be a problem, as demonstrated by Ross and Altman in [41], where wind tunnel testing in Savonius rotors has

shown that blockage can greatly affect wind tunnel testing, especially in high-solidity Savonius rotors.

Since VAWTs are well-suited to low power, off-grid electricity production, there are also concerns about blade icing when turbines are installed in cold, moist regions, where grid power may not be available. Blade icing has been well-researched in the HAWT setting, however very little has been done in comparison with vertical axis turbines. Li et al. [60] investigate leading edge icing on straight-bladed VAWTs, and determine through experimental means, using clay as an ice simulator, that the more ice accumulates on the rotor, the more the power coefficient of the VAWT is lowered. Li et al. [61] perform another work where icing and ice accretions on airfoils at some typical attack angles are simulating using a water spray nozzle. This nozzle is inserted into a wind tunnel with sub-zero wind temperatures, which contains a static VAWT airfoil blade. The study finds that icing characteristics and hence, VAWT performance in cold/moist conditions, is shown to be heavily dependent on attack angle and air moisture composition. Numerical and experimental results in [62] confirm that icing does in fact lead to decreasing dynamic torque coefficients, and thus, decreasing efficiency.

Utilizing commercial software, Howell et al [42] investigate rotors with two and three straight blades, utilizing both 2-D and 3-D simulations, as well as experimental techniques. The experiment is conducted analyzing both the two and three-bladed rotors utilizing a NACA 0022 airfoil blade type. The 3-D simulations are shown to be much more accurate than the 2-D approach in predicting torque at varying tip-speed ratios with constant wind velocity. The results from experiments also conclude that the three-bladed rotor with high solidity achieves much better performance than the two-bladed rotor.

Many experimental tests in the recent literature are coupled with power and torque control in VAWTs. Through carefully controlling rotational speed and utilizing sophisticated gearbox/generator designs, tip-speed ratios can be kept near optimal values for a range of wind speeds. While speed and/or torque control is not the subject of this review, nevertheless these experimental studies, for example in [57], provide useful insight into actual performance of lift-based VAWT rotors in a variety of wind and rotor speeds.

Larger scale wind tunnel tests are also in the works, with geometric scaling possible for larger capacity VAWT systems. For example, Battisti et al [43] test a three-bladed Darrieus rotor in a large scale wind tunnel to investigate performance. This study, as well as others like it, allow for VAWT designers without experimental means to validate CFD or numerical codes, and thus facilitate more research into designing new blade shapes and rotors for optimal performance. Bravo et al [52] utilize a 9 m wind tunnel to thoroughly test a 3.5 kW VAWT. Similar to many other smaller scale VAWTs, the optimal tip-speed ratio was found to be around 1.6 for operational wind speeds of between 5 and 15 m/s, where a maximum power coefficient was found to be 0.3.

Although wind tunnel testing can be a viable way to both validate numerical simulations and perhaps to test small-scale rotors, larger wind tunnels are generally unavailable for testing of VAWT rotors above 10 kW. Large field testing of rotors is becoming more common, for example in Uppsala University in Sweden, where a 12 kW H-Darrieus type VAWT has been constructed and tested [44]. The turbine has a height of 6 m and a blade radius of 3 m, and is operated outdoors in a range of wind speeds and tip-speed ratios. Among other findings, the maximum power coefficient found was 0.29 corresponding to a tip-speed ratio of 3.3.

Due to the recent large market penetration of the HAWT into offshore applications, some have proposed that the VAWT may be suitable for the same application. However, as mentioned previously, the reason why VAWTs have not been scaled to similar capacities as

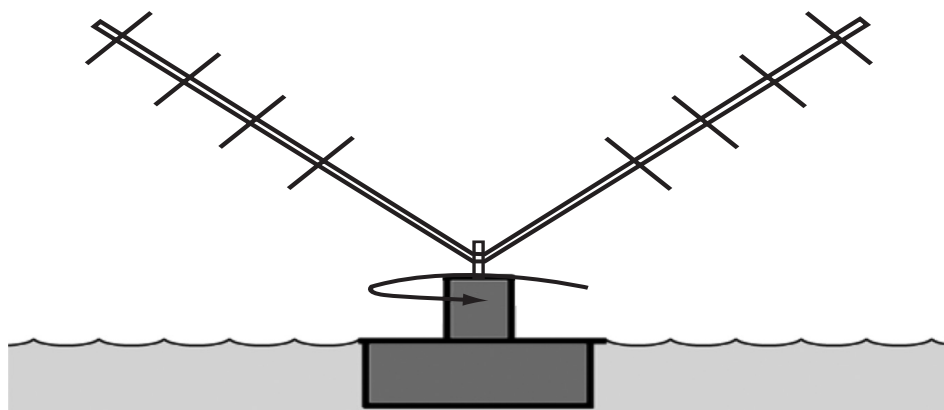


Figure 9: An offshore V-shaped VAWT with sails [modified from 45].

HAWTs is due to the large, oscillatory stresses realized by the tower and rotor arms, making high capacity VAWTs in general difficult to safely operate. However, Blusseau and Patel [59] argue that VAWTs may be more suitable to offshore applications than HAWTs, due to low center of gravity and base location of generating machinery. This study addresses the gyroscopic effect, one of the more significant problems with offshore VAWT design, for a VAWT mounted on semi-submersible platforms. The turbine they consider is the NOVA turbine, which is investigated in Kolios et al [45]. Here, the novel offshore VAWT concept is envisioned, which could be used in larger offshore designs. They attempt to alleviate the problem of large oscillating tower moment by using a V-shaped rotor fitted with smaller sails (Fig. 9). By altering the sails on the rotor, significant reductions in overturning moment can be realized, allowing for greater stability of the VAWT.

Another possible solution to the problem of large bending moment in offshore VAWT designs is proposed by Akimoto et al. [63], where a floater, capable of rotating with the turbine to allow for heaving, is designed to help minimize costs associated with offshore VAWTs. By allowing a larger tilt angle, the total weight of the system is reduced, and their preliminary estimates indicate that float systems such as these increase economic viability.

In yet another example of continuing efforts of incorporating VAWT rotors into offshore wind energy, DTU researchers in Denmark [68] are investigating a floating Darrieus Egg-Beater type turbine, which includes the possibility of floating generators which rotate with the turbine. If proven plausible and reliable, these and other promising offshore designs (for example the floating VAWT studied by Vita et al [46]), can help VAWT turbines enjoy significant gains in offshore applications.

4. CONCLUSIONS

A thorough review of recent advances in Vertical Axis Wind Turbine (VAWT) rotor design is presented, with emphasis on scholarly publications since 2005. Although most wind turbines in use today are of the horizontal-axis type, much attention has been devoted to research and development of VAWT rotors for small scale applications, including urban areas, off-grid production and other specialized applications.

Generally, VAWTs are less efficient than HAWTs. However, the increased recent interest in VAWTs is due to several factors. First and foremost, the comfort of placing the generator and gearbox on the ground reduces its maintenance when compared to HAWTs. The tall towers of HAWTs are expensive to install, and often difficult to transport. Furthermore, since VAWTs

operate virtually independent of wind direction, they do not need to be actively oriented to the direction of the wind, which eliminates the need for a yaw control system - making the aggregate WEC device more complex, i.e. less reliable. This is negated by the fact that most wind turbine sites do not exhibit frequent changes of wind direction. Because of this inherent independence of wind direction, VAWT designs are likely better for situations where wind velocities are relatively inconsistent.

VAWTs do not furl, i.e., do not have the ability to yaw out of the wind direction to reduce output in high load situations. Since the power output is proportional to the cube of the wind velocity, doubling the wind velocity raises the potential power extracted by a factor of eight. Designing VAWTs to withstand such extreme wind power and extracting the "excess" power raises the cost of implementation. The cycle of lift and loss on the same blade during each rotation also adds fatigue. Fatigue also results from the turbulence downwind created by the upwind blade of the same configuration.

In order to determine VAWT feasibility in inconsistent wind areas, many recent studies have been completed in order to increase performance. Most are concerned with lift-based, Darrieus-type rotors, with drag-based rotors and hybrid designs also receiving attention. The review of the most recent works in this area indicates a few notable conclusions, which can be summarized as follows:

- Blade pitching techniques have been known to increase the performance of both HAWTs and VAWTs, and are either passive or active in nature. Active blade pitching involves a control system and actuators, and detracts from the initial low-cost attractiveness of VAWTs. Passive techniques are becoming more prevalent, with a few recent works studying novel designs and mechanisms used to tilt blades as they revolve around the vertical axis. While passive approaches have enjoyed limited success, active strategies have shown significant improvement can be made by blade pitching techniques with up to 30% increases in power production possible.
- Self-starting of lift-based VAWTs is an attractive feature, with many recent studies concerned with enhancing the self-starting nature using such techniques as hinged flaps, asymmetric airfoils and multi-stage turbines. CFD techniques are also employed to assess self-starting. However some claims have been made which question the accuracy of current modeling techniques in predicting the conditions in which self-starting occurs. While certain designs do enhance the range of self-starting conditions, they come at an added design cost or decreased operational efficiency.
- Though typically lower in power generation, drag based VAWTs usually self-start under much lower wind speeds, and can provide high torque for small power production. Most recent works involving drag-based VAWTs investigate possible performance gains using venting, guide vanes and apertures in two or three bucket systems, with most studies concluding the two-bucket rotor is generally superior to the three-bucket rotor.
- Hybrid VAWT rotors are also a popular area of interest, where most commonly a lift-based rotor is fitted with a second drag-based rotor of smaller armature radius on the same axis. This allows the high-torque drag based rotor, which has better self-starting characteristics, to increase operational range of the combined VAWT design and hence, increases average power coefficient. Studies in this subject have

concluded that these hybrid designs do indeed enjoy much higher power production over a more realistic and variable range of wind conditions.

Based on the above challenges, we envision four immediate research needs for VAWTs:

- Affordable output control in high winds, in order to compete better with HAWTs. This includes blade pitching and torque control techniques;
- Better design to significantly reduce fatigue and vibration due to oscillatory loading of the support and blades;
- Improved conversion efficiency - in lift based systems, this is accomplished by reducing drag losses;
- Improved self-starting characteristics of lift-based (Darrieus) type rotors. If self-starting can be achieved at relatively low wind speeds, overall conversion efficiency is greatly enhanced.

Though the activity involving research and development of VAWT rotor design is increasing, it is expected to continue this trend in the future. Global energy demand is sure to follow the same trends, and energy capture using WEC systems, including VAWT designs, must continue to flourish and become one of many solutions to fit energy supply with demand worldwide.

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